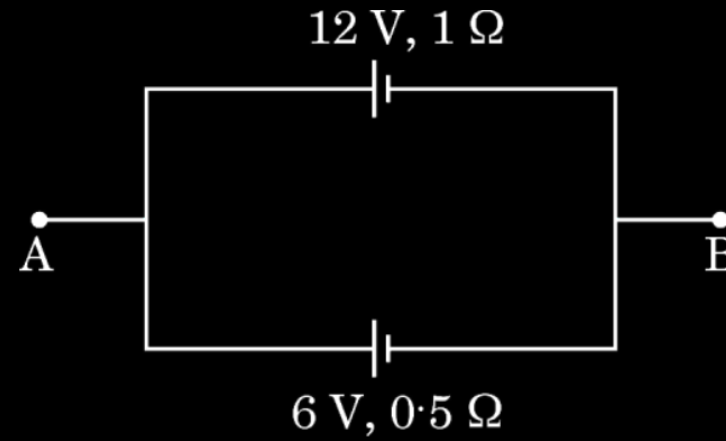
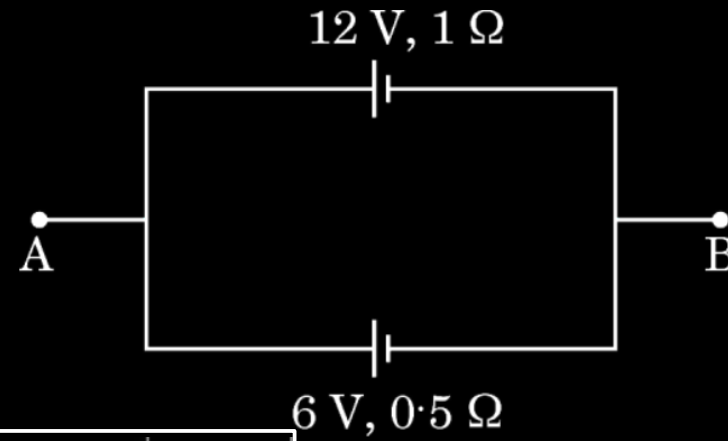


- (ii) Two cells of emfs 12 V and 6 V are connected in parallel as shown in the figure. Their internal resistances are $1\ \Omega$ and $0.5\ \Omega$ respectively. Calculate the emf and internal resistance of the equivalent cell between points A and B.



- (ii) Two cells of emfs 12 V and 6 V are connected in parallel as shown in the figure. Their internal resistances are $1\ \Omega$ and $0.5\ \Omega$ respectively. Calculate the emf and internal resistance of the equivalent cell between points A and B.



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$$\begin{aligned} \text{(ii) } \varepsilon_{AB} &= \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} \\ &= \frac{12 \times 0.5 + 6 \times 1}{1.5} \end{aligned}$$

$$\begin{aligned} \text{Internal resistance} &= \frac{r_1 r_2}{r_1 + r_2} \\ &= \frac{0.5 \times 1}{1.5} \\ &= \frac{1}{3} \Omega \end{aligned}$$

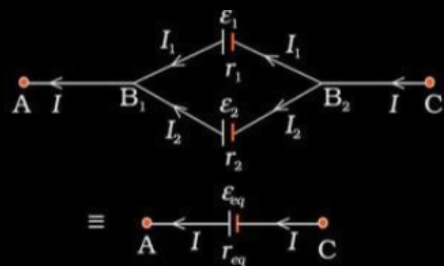
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (a) Two cells of emf E_1 and E_2 with internal resistances r_1 and r_2 respectively, are connected in parallel by connecting their positive terminals together and negative terminals together. Deduce an expression for equivalent emf and equivalent internal resistance of the combination.
- (b) A parallel combination, as stated in (a) above, of two cells of emfs E and $3E$ and internal resistances R each is connected across a resistance $2R$. Find the current that flows through resistance $2R$.



Let I_1 and I_2 are the currents leaving from the positive electrodes of the cells ε_1 and ε_2 respectively, Hence $I = I_1 + I_2$

Potential difference across the terminals of cell ε_1 is

$$V = \varepsilon_1 - I_1 r_1$$

Potential difference across the terminals of cell ε_2 is

$$V = \varepsilon_2 - I_2 r_2$$

$$I = I_1 + I_2$$

$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} \right) - V \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \left(\frac{r_1 r_2}{r_1 + r_2} \right)$$

$$V = \varepsilon_{eq} - I r_{eq}$$

$$\varepsilon_{eq} = \left(\frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} \right)$$

$$r_{eq} = \left(\frac{r_1 r_2}{r_1 + r_2} \right)$$

$\frac{1}{2}$

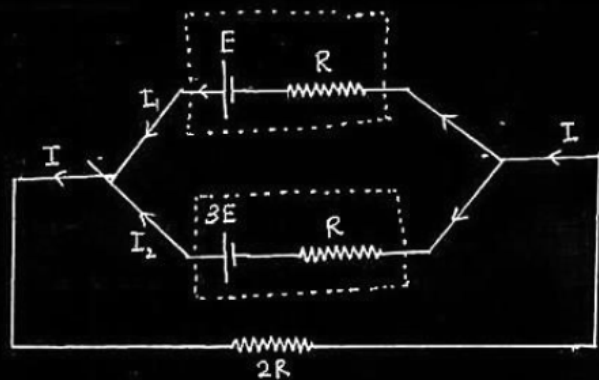
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

(b)



Equivalent emf

$$E_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$E_{eq} = \frac{E \times R + 3E \times R}{R + R}$$

$$E_{eq} = \frac{4ER}{2R}$$

 $\frac{1}{2}$

$$E_{eq} = 2E$$

Equivalent resistance

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

$$r_{eq} = \frac{RR}{R + R}$$

$$r_{eq} = \frac{R}{2}$$

$$I = \frac{E_{eq}}{2R + \frac{R}{2}}$$

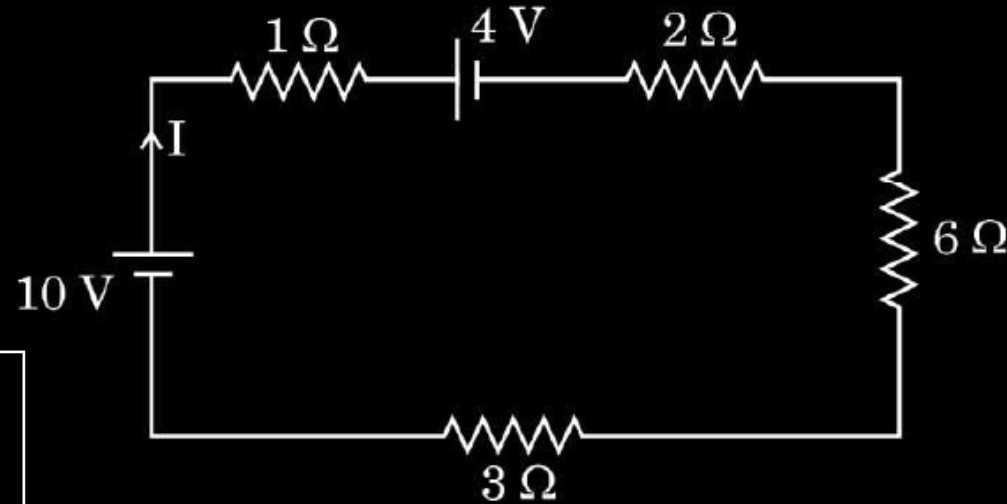
$$I = \frac{4E}{5R} A$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

OR

- (b) (i) Define electrical conductivity of a wire. Give its SI unit.
- (ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?
- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.

5



(i) Definition & S.I. Unit	1½
(ii) Explanation of internal resistance for low voltage and high voltage battery	1+1
(iii) Total energy	1½

(i) Definition : Electrical conductivity is defined as the measure of a material's ability to carry a current through it. Alternatively: It is the reciprocal of the resistivity. Alternatively: It is defined as the current density per unit electric field.	1
S.I. Unit: $(\text{ohm})^{-1}\text{-m}^{-1}$ or S-m^{-1}	$\frac{1}{2}$
(ii) Low voltage Battery- Internal resistance should be low.	1
High voltage Battery – Internal resistance should be high.	1
(iii) Applying Kirchhoff's loop rule $10 - I \times 1 - 4 - 2I - 6I - 3I = 0$ $12I = 6 \Rightarrow I = 0.5 \text{ A}$ Heat energy $H = I^2 R t$ $H = 0.25 \times 12 \times 60 = 180J$	$\frac{1}{2}$
	$\frac{1}{2}$

25. Two identical cells, each of emf E and internal resistance r , are connected with a load resistance R , first in series and then in parallel. Obtain the condition under which the current through R is same in both cases. 2

Calculation of current for cells in series and parallel $\frac{1}{2} + \frac{1}{2}$
Obtaining the condition when current is the same in both the cases 1

In series,

$$\varepsilon_{\text{eq}} = \varepsilon + \varepsilon = 2\varepsilon$$

$$I_s = \frac{2\varepsilon}{2r + R}$$

In parallel,

$$\varepsilon_{\text{eq}} = \varepsilon$$

$$r_{\text{eq}} = \frac{r}{2}$$

$$I_p = \frac{\varepsilon}{\frac{r}{2} + R}$$

$$\therefore I_p = I_s$$

$$\frac{2\varepsilon}{2r + R} = \frac{\varepsilon}{\frac{r}{2} + R}$$

$$\frac{2}{2r + R} = \frac{1}{\frac{r}{2} + R}$$

$$2\frac{r}{2} + 2R = 2r + R$$

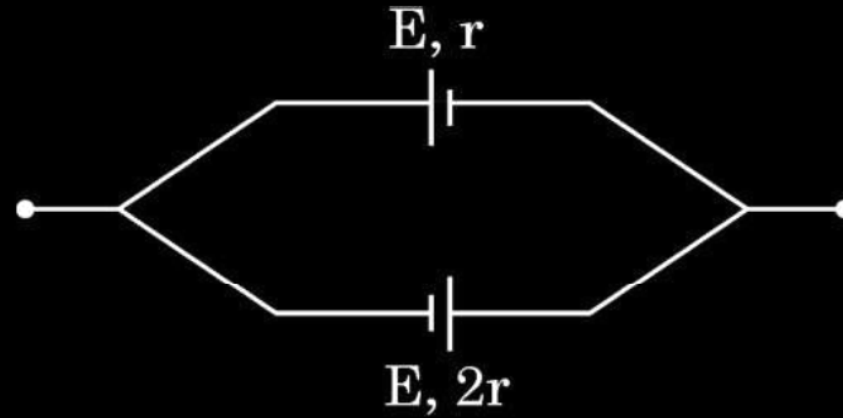
$$r = R$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

14. Two cells of emf E each and internal resistances r and $2r$ are connected in parallel as shown in the figure. The equivalent emf of the combination is :



- (a) zero
- (b) $\frac{E}{2}$
- (c) $\frac{E}{3}$
- (d) E

29. Two cells of emf E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their terminals of the same polarity connected together. Obtain an expression for the equivalent emf of the combination. 3



$$\frac{1}{2}$$

$$V \equiv V(B_1) - V(B_2) = \varepsilon_1 - I_1 r_1$$

$$\frac{1}{2}$$

$$V \equiv V(B_1) - V(B_2) = \varepsilon_2 - I_2 r_2$$

$$\frac{1}{2}$$

$$I = I_1 + I_2$$

$$\frac{1}{2}$$

$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} \right) - V \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$$

$$\frac{1}{2}$$

$$\therefore V = \varepsilon_{eq} - I r_{eq}$$

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

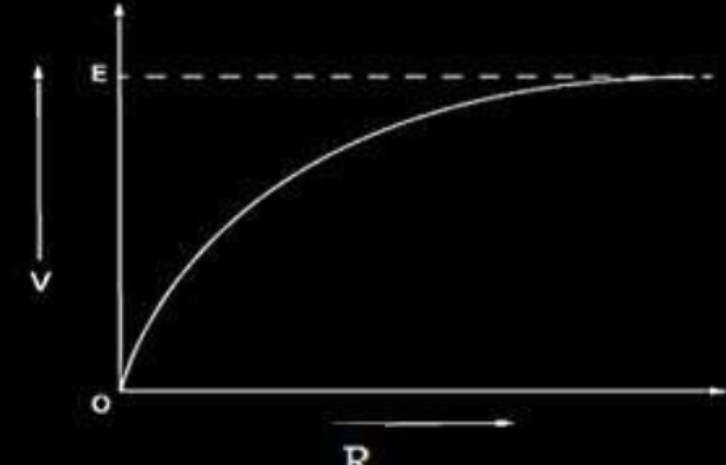
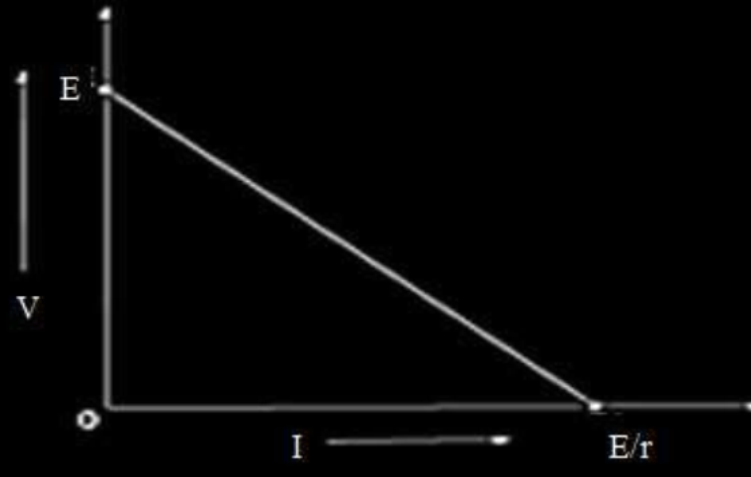
$$\frac{1}{2}$$

OR

- (b) (i) A cell emf of (E) and internal resistance (r) is connected across a variable load resistance (R). Draw plots showing the variation of terminal voltage V with (i) R and (ii) the current (I) in the load. 2
- (ii) Three cells, each of emf E but internal resistances $2r$, $3r$ and $6r$ are connected in parallel across a resistor R .

Obtain expressions for (i) current flowing in the circuit, and (ii) the terminal potential difference across the equivalent cell. 3

- | | |
|---|------------------|
| (i) | |
| i Plot showing variation of V with R | (1) |
| ii Plot showing variation of V with I | (1) |
| (ii) | |
| (i) Obtaining expression for current flowing through circuit | $(1\frac{1}{2})$ |
| (ii) Obtaining expression for terminal potential difference across the equivalent cell. | $(1\frac{1}{2})$ |

(i) i. 	(ii) $\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$ $\frac{1}{r_{eq}} = \frac{1}{2r} + \frac{1}{3r} + \frac{1}{6r}$ $r_{eq} = r$ Given cells are of equal emf (E) and connected in parallel, so $\frac{E_{eq}}{r_{eq}} = \frac{E}{2r} + \frac{E}{3r} + \frac{E}{6r}$ $E_{eq} = E$ Current flowing $I = \frac{E_{eq}}{R + r_{eq}}$ $I = \frac{E}{R + r}$ The terminal potential difference $V = E_{eq} - Ir_{eq}$ $V = E - \frac{E}{(R + r)} \times r$ $V = \frac{ER}{R + r}$
ii. 	$E_{eq} = E$ Current flowing $I = \frac{E_{eq}}{R + r_{eq}}$ $I = \frac{E}{R + r}$ The terminal potential difference $V = E_{eq} - Ir_{eq}$ $V = E - \frac{E}{(R + r)} \times r$ $V = \frac{ER}{R + r}$

4. Two cells, one of emf $2E$ and internal resistance r and the other of emf E and internal resistance $r/2$ are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a) $3E$

(b) $\frac{4E}{3}$

(c) $2E$

(d) $\frac{3E}{4}$

4. Two cells, one of emf $2E$ and internal resistance r and the other of emf E and internal resistance $r/2$ are connected in parallel, by connecting their positive terminals together and their negative terminals together. The equivalent emf of the combination is :

(a) $3E$

(b) $\frac{4E}{3}$

(c) $2E$

(d) $\frac{3E}{4}$

(b) $\frac{4E}{3}$

(ii) Two cells of emfs 2.0 V and 6.0 V and internal resistances 0.1Ω and 0.4Ω respectively, are connected in parallel. The equivalent emf of the combination will be :

(A) 2.0 V

(B) 2.8 V

(C) 6.0 V

(D) 8.0 V

(ii) Two cells of emfs 2.0 V and 6.0 V and internal resistances $0.1\ \Omega$ and $0.4\ \Omega$ respectively, are connected in parallel. The equivalent emf of the combination will be :

(A) 2.0 V

(B) 2.8 V

(C) 6.0 V

(D) 8.0 V

(iv) (a) Five identical cells, each of emf 2 V and internal resistance $0.1 \, \Omega$ are connected in parallel. This combination in turn is connected to an external resistor of $9.98 \, \Omega$. The current flowing through the resistor is :

(A) $0.05 \, \text{A}$

(B) $0.1 \, \text{A}$

(C) $0.15 \, \text{A}$

(D) $0.2 \, \text{A}$

(iv) (a) Five identical cells, each of emf 2 V and internal resistance $0.1 \, \Omega$ are connected in parallel. This combination in turn is connected to an external resistor of $9.98 \, \Omega$. The current flowing through the resistor is :

(A) $0.05 \, \text{A}$

(B) $0.1 \, \text{A}$

(C) $0.15 \, \text{A}$

(D) $0.2 \, \text{A}$

3. A student is asked to connect four cells, each of emf E and internal resistance r , in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

(A) $4E$ and $2r$

(B) $4E$ and $3r$

(C) $3E$ and $4r$

(D) $2E$ and $4r$

3. A student is asked to connect four cells, each of emf E and internal resistance r , in series. But she/he connects one cell wrongly in series with the other cells. The equivalent emf and the equivalent internal resistance of the combination will be :

(A) $4E$ and $2r$

(B) $4E$ and $3r$

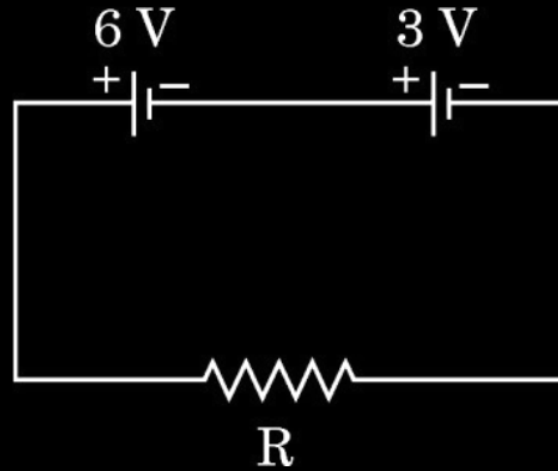
(C) $3E$ and $4r$

(D) $2E$ and $4r$

(D) $2E$ and $4r$

17. Two batteries of emfs 6 V and 3 V and internal resistances $0.8\ \Omega$ and $0.2\ \Omega$ respectively are connected in series to an external resistance R , as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.

2



Finding the value of R

2

$$i = \frac{9}{R+1}$$

As potential difference across 6V is zero;

$$6 - ir = 0 \Rightarrow 6 - \left(\frac{9}{R+1}\right)(0.8) = 0$$

On solving;

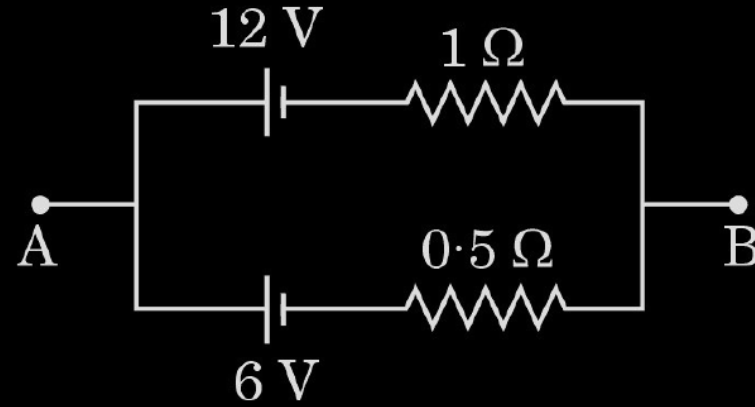
$$R = 0.2\Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

1

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



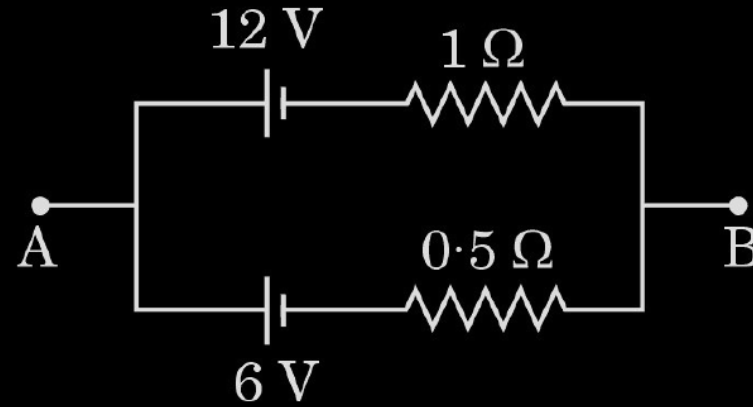
(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

(B) 8V

17. Two identical cells, each of emf E and internal resistance r , are connected in series with a resistor of $8\ \Omega$. A current of 1.5 A flows in the circuit. When these cells are connected in parallel, they send 1.0 A current through the same resistor. Calculate the internal resistance, r and emf, E of the cell. 2

Calculation of (i) Internal Resistance

1

(ii) EMF

1

In Series Combination

$$\begin{aligned} 2E &= I(2r+8) \\ &= 1.5(2r+8) \dots\dots\dots(1) \end{aligned}$$

In Parallel Combination

$$\begin{aligned} E &= I\left(\frac{r}{2} + 8\right) \\ E &= 1\left(\frac{r}{2} + 8\right) \dots\dots\dots(2) \end{aligned}$$

Solving equation (1) & (2)

$$2 = \frac{1.5(2r+8)}{\left(\frac{r}{2} + 8\right)}$$

$$r + 16 = 3r + 12$$

$$2r = 4$$

$$r = 2\Omega$$

Substituting the value of r in equation(2)

$$E = \left(\frac{2}{2} + 8\right)$$

$$E = 9V$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(b) Obtain an expression for emf and internal resistance of a cell which is equivalent to parallel combination of two cells of emf and internal resistance (E_1, r_1) and (E_2, r_2) , respectively.

3

Obtaining expression for equivalent emf	2
Equivalent resistance	1

When two cells are connected in parallel the total current

$$I = I_1 + I_2 \text{ -----(1)}$$

$$I = \frac{\varepsilon_1 - V}{r_1} + \frac{\varepsilon_2 - V}{r_2}$$

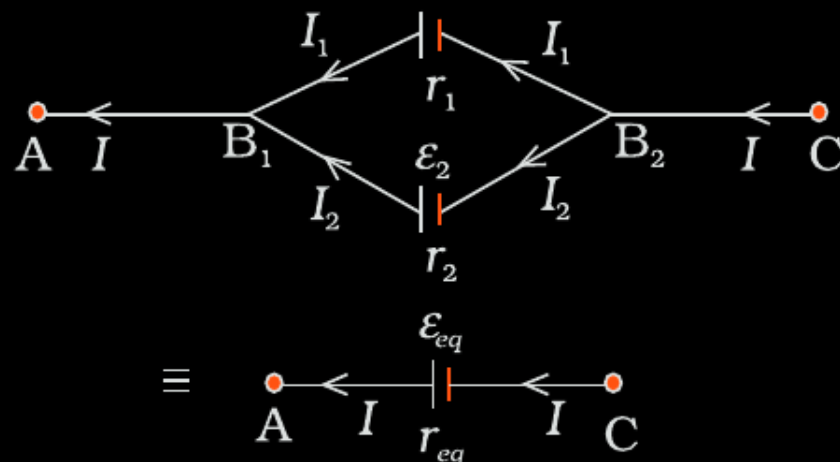
$$I = \left(\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} \right) - V \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \left(\frac{r_1 r_2}{r_1 + r_2} \right)$$

Hence

$$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$



1/2

1/2

1/2

1/2

1/2

1/2

17. n identical cells, each of e.m.f. E and internal resistance r , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'.

17. n identical cells, each of e.m.f. E and internal resistance r , are connected in series. Later on it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. 2

Calculating the potential difference	2
Net e.m.f = $(n-4)E$ $\therefore I = \frac{(n-4)E}{nr}$	$\frac{1}{2}$
Potential difference across 'X' $V = E + Ir$	$\frac{1}{2}$

22. (a) Two batteries of emf's 3V & 6V and internal resistances $0.2\ \Omega$ & $0.4\ \Omega$ are connected in parallel. This combination is connected to a $4\ \Omega$ resistor. Find :

- (i) the equivalent emf of the combination
- (ii) the equivalent internal resistance of the combination
- (iii) the current drawn from the combination

OR

22. (a) Two batteries of emf's 3V & 6V and internal resistances $0.2\ \Omega$ & $0.4\ \Omega$ are connected in parallel. This combination is connected to a $4\ \Omega$ resistor. Find :

- (i) the equivalent emf of the combination
- (ii) the equivalent internal resistance of the combination
- (iii) the current drawn from the combination

(i) Because $E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$	$\frac{1}{2}$
$E_{eq} = \frac{3 \times 0.4 + 6 \times 0.2}{0.6} = 4\text{ V}$	$\frac{1}{2}$
(ii) $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$	$\frac{1}{2}$
$r_{eq} = \frac{0.2 \times 0.4}{0.2 + 0.4} = 0.133\Omega$	$\frac{1}{2}$
(iii) $I = \frac{E}{R + r_{eq}}$	$\frac{1}{2}$
$I = \frac{4}{4 + 0.13} = \frac{4}{4.13}\text{ A}$	$\frac{1}{2}$
$I = 0.9\text{ A}$	

28. (a) Two cells of emf E_1 and E_2 have their internal resistances r_1 and r_2 , respectively. Deduce an expression for the equivalent emf and internal resistance of their parallel combination when connected across an external resistance R . Assume that the two cells are supporting each other.
- (b) In case the two cells are identical, each of emf $E = 5 \text{ V}$ and internal resistance $r = 2 \Omega$, calculate the voltage across the external resistance $R = 10 \Omega$.

3

a) Internal resistance	1 ½ mark
b) Voltage across R	1 ½ mark

Current drawn from cell -1

$$I_1 = \frac{E_1 - V}{r_1}$$

Current drawn from cell -2

$$I_2 = \frac{E_2 - V}{r_2}$$

Resultant current

$$I = I_1 + I_2$$

On solving

$$\therefore I = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - V \left(\frac{r_2 + r_1}{r_1 r_2} \right)$$

$$\therefore V = \frac{E_1 r_2 + E_2 r_1}{r_1 r_2} - I \left(\frac{r_1 r_2}{r_2 + r_1} \right)$$

$$V = E_{eq} - I r_{eq}$$

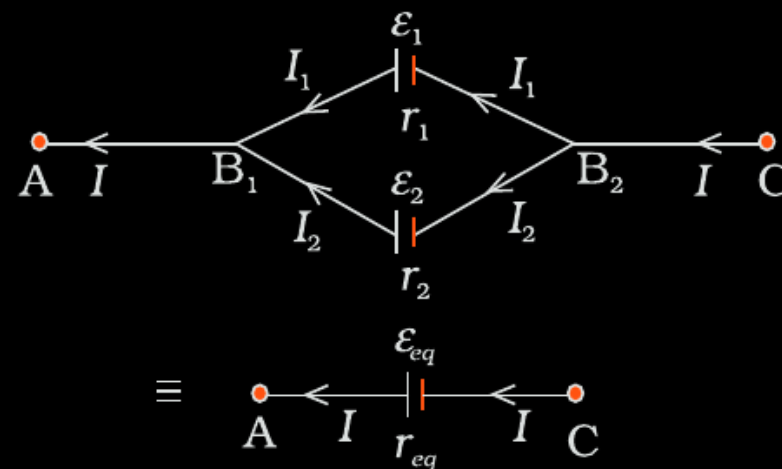
$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_2 + r_1}$$

1/2

1/2

1/2



Current through R

$$r_{eff} = \frac{r_1 r_2}{r_1 + r_2} = \frac{2 \times 2}{2 + 2} = 1 \Omega$$

$$I = \frac{E_{effect}}{R + r_{eff}} = \frac{5}{10 + 1} = \frac{5}{11} A$$

P.D across R

$$= \frac{5}{11} \times 10 = 4.54 \text{ volt}$$

1/2

1/2

1/2

- (a) Two cells of different emfs and internal resistances are connected in parallel with one another. Derive the expression for the equivalent emf and equivalent internal resistance of the combination.
- (b) Two identical cells of emf 1.5 V and internal resistance r are each connected in parallel providing a supply to an external circuit consisting of two resistances of $17\ \Omega$ each joined in parallel. A very high resistance voltmeter reads the terminal voltage of the cell to be 1.4 V. Calculate the internal resistance of each cell.

a)	Expression of current in terms of E and V	1
	Expression of voltage in terms of emf and internal resistance	1
	Expression for E_{eq} and r_{eq}	$\frac{1}{2} + \frac{1}{2}$
b)	Value of current & value of internal resistance	1+1

$$I = I_1 + I_2$$

Potential difference across $B_1 B_2$

$$V = E_1 - I_1 r_1 \quad \Rightarrow \quad I_1 = \frac{E_1 - V}{r_1}$$

$$V = E_2 - I_2 r_2 \quad \Rightarrow \quad I_2 = \frac{E_2 - V}{r_2}$$

$$I = I_1 + I_2$$

$$= \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$$

$$= \left(\frac{E_1}{r_1} + \frac{E_2}{r_2} \right) - V \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$$

$$V = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} - \frac{I r_1 r_2}{r_1 + r_2}$$

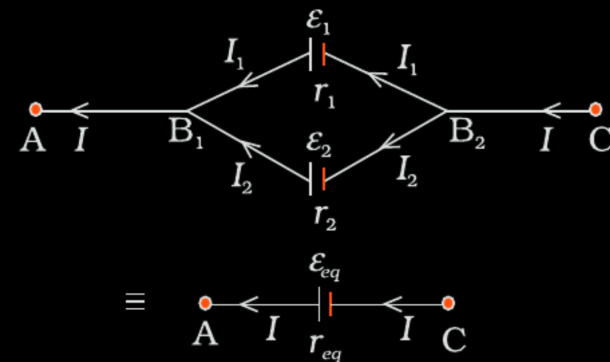
Compare with

$$V = E_{eq} - I r_{eq}$$

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$$

$\frac{1}{2}$

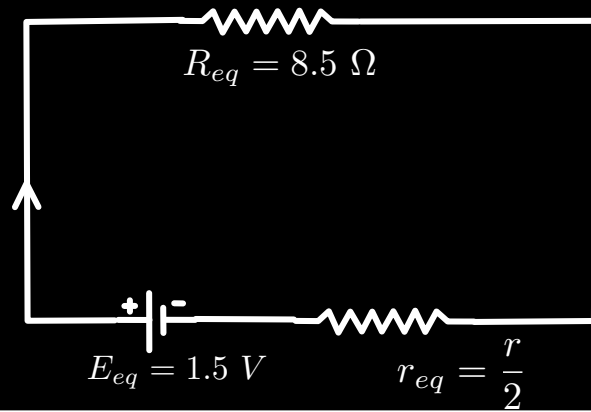
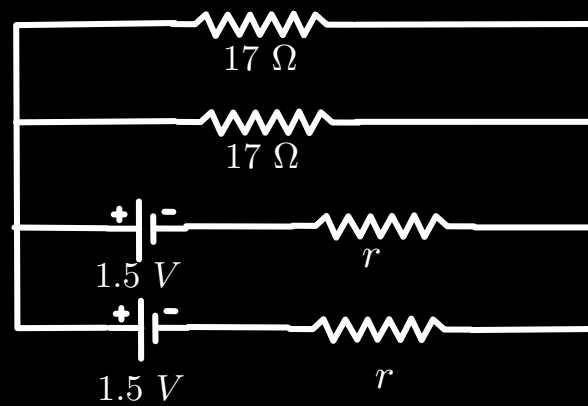


$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



Equivalent circuit

1

$$I = \frac{V}{R_{eq}} = \frac{1.4}{8.5}\ A$$

And

$$V = E_{eq} - Ir_{eq}$$

$$\Rightarrow 1.4 = 1.5 - \frac{1.4}{8.5} \times \frac{r}{2}$$

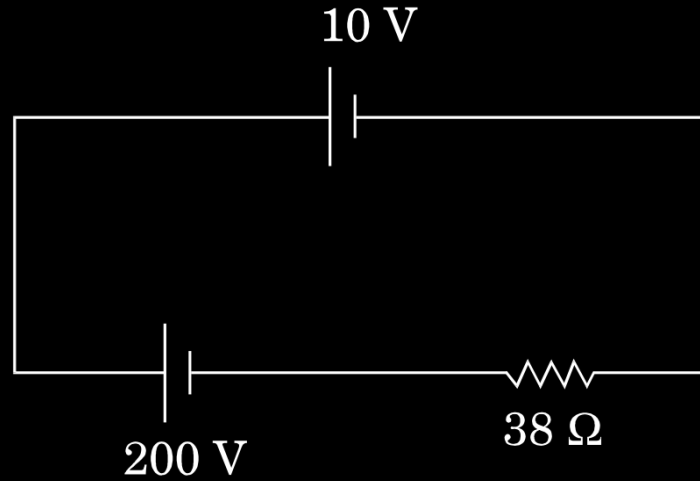
$$\Rightarrow r = 1.21\ \Omega$$

1

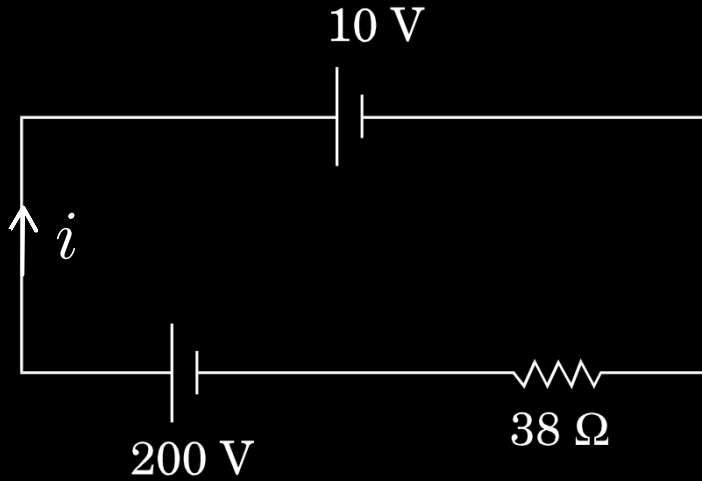
Two cells of emfs ε_1 & ε_2 and internal resistances r_1 & r_2 respectively are connected in parallel. Obtain expressions for the equivalent.

- (i) resistance and
- (ii) emf of the combination

A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



By Kirchhoff's law, we have, for the loop ABCD,

$$+200 - 38i - 10 = 0$$

$$\therefore i = \frac{190}{38} \text{ A} = 5 \text{ A}$$

1

1

Two cells of emfs 1.5 V and 2.0 V having internal resistances $0.2\ \Omega$ and $0.3\ \Omega$ respectively are connected in parallel. Calculate the emf and internal resistance of the equivalent cell.

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$$\text{emf} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

$$= \frac{1.5 \times 0.3 + 2 \times 0.2}{0.2 + 0.3} \text{ V}$$

$$= \frac{0.45 + 0.40}{0.5} \text{ V} = 1.7 \text{ V}$$

$$r = \frac{r_1 r_2}{r_1 + r_2}$$

$$= \frac{0.2 \times 0.3}{0.2 + 0.3} \Omega$$

$$= \frac{0.06}{0.5} \Omega$$

$$= 0.12 \Omega$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

Two identical cells of emf 1.5 V each joined in parallel supply energy to an external circuit consisting of two resistances of $7\ \Omega$ each joined in parallel. A very high resistance voltmeter reads the terminal voltage of cells to be 1.4 V . Calculate the internal resistance of each cell.

We have, for a single cell,

$$r = \left(\frac{E}{V} - 1 \right) R$$

$\therefore$ For the parallel combination, as given in the question,

$$\frac{r}{2} = \left(\frac{E}{V} - 1 \right) \frac{R}{2}$$

$$\therefore r = \left(\frac{1.5}{1.4} - 1 \right) \times 7 \Omega$$

$$= \frac{0.1}{1.4} \times 7 \Omega$$

$$= 0.5 \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(Alternatively,

$$I = \frac{V}{(R/2)}$$

$$\text{And } E = V - I(r/2)$$

$$\text{This gives } I = \frac{1.4}{7/2} \text{ A} = 0.4 \text{ A}$$

$$\therefore \frac{r}{2} = \frac{1.5 - 1.4}{0.4} = 0.25$$

$$\therefore r = 0.5 \Omega$$

(Note: If the student just draws the circuit diagram of the setup but does not do any calculations, award 1 mark only.)

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

